Scaled citizen science participatory variety selection provides socially inclusive feedback to the cassava breeding in Nigeria and opens the way to empower women and men crop users within the breeding process.



Béla Teeken *et al.*
Gender conference New Delhi
9 to 12 October 2023

Outline

Context

- Towards demand driven and socially/gender inclusive cassava breeding in Nigeria

Methodology

- Setup of participatory variety selection trials in Nigeria
- Principles of Tricot trials
- Selecting participants

Results

- Trait & variety preferences of tricot farmers and their processors and relevant social covariates and their intersection

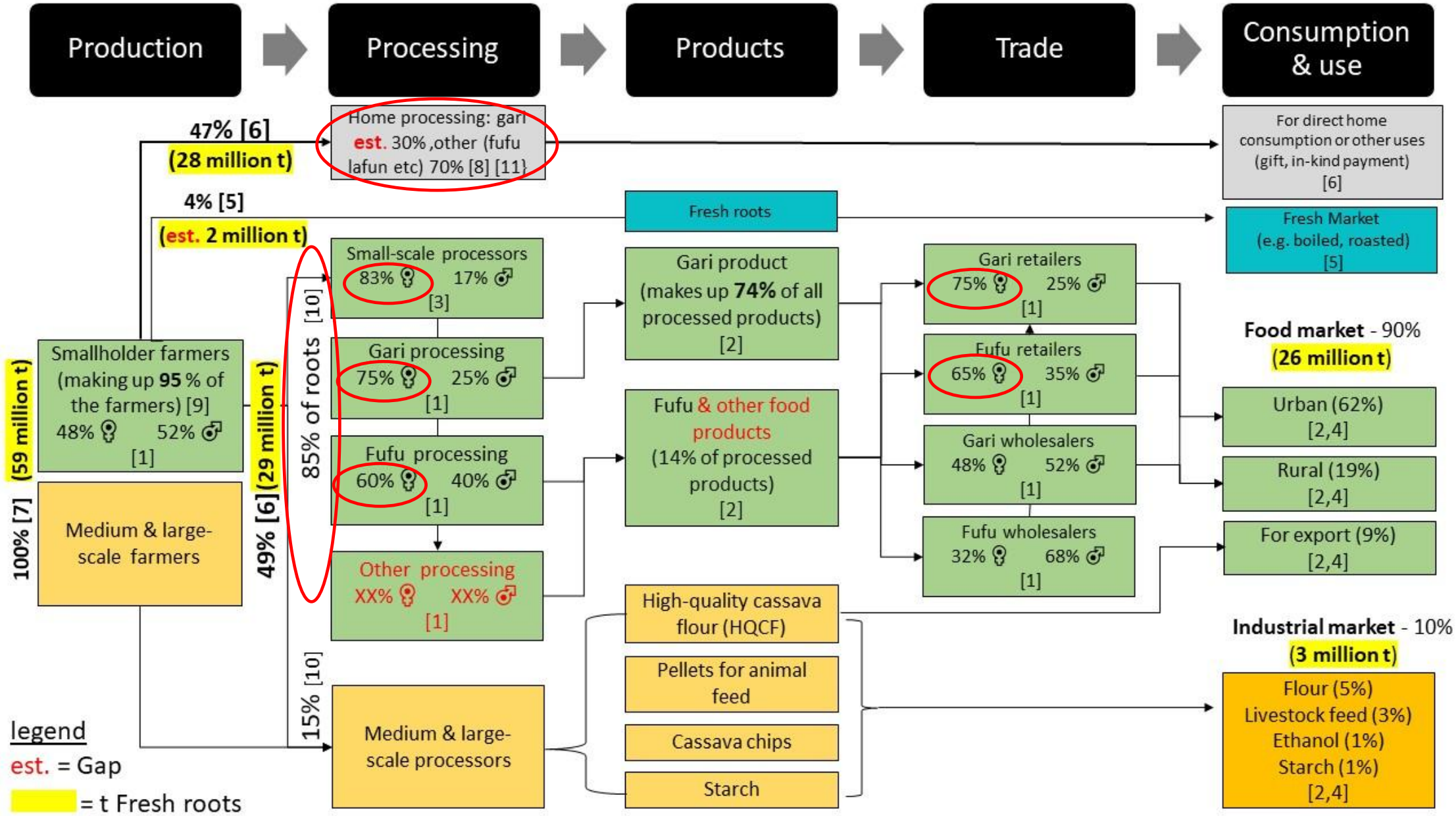
Conclusion

- Sustainably integrating tricot in breeding as improved on farm testing for variety release.
  - Social covariate (automated) analysis standardly integrated within breeding data
  - Inclusion of crop users as stakeholders in the breeding process

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Women largely dominate in food processing & marketing

They determine varieties suitable for the whole value chain



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State representative survey, participatory processing and consumer testing on preferred characteristics

(Alamu et al. 2023; Olaosebikan *et al*. 2023, Chijioke 2020, Ndjouenkeu, 2020 Teeken *et al*. 2018,)



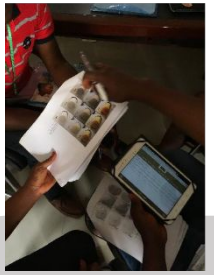
Gender responsive mother baby trial approach with farmers and processors

(Teeken *et al.* 2021)



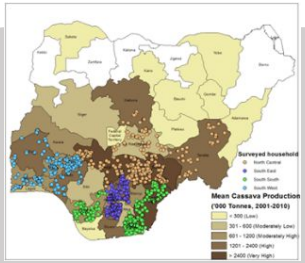
TRICOT citizen science trials: upscaling baby trial approach with farmers and processors

(Method: van Etten *et al*. 2016)




1000 minds study: economic value of agronomic and food product quality traits

(Balogun *et al*. (2022)



IITA Cassava Monitoring Study on adoption varieties in Nigeria (2500 households)

(Wossen *et al*. 2017)

RESEARH THAT INFORMED THE CURRENT BREEDING PRODUCT PROFILE: PROCESSED CASSAVA FOOD (GARI AND FUFU)



Demand Creation Trials

(ThankGod et al.,2023)

Day 2

Olamide Olaosebikan: Using the **G+** systematic customer and breeding product profiling tools to validate gender equitable breeding in Nigeria **Session 2.3B ·(3:50h- 5:00h)**

Tricot citizen science trials: upscaling baby trial approach in Nigeria (need for larger representativeness)

Tricot trials in Nigeria (2019-2020 and 2021-2022 seasons, NRCRI / IITA)

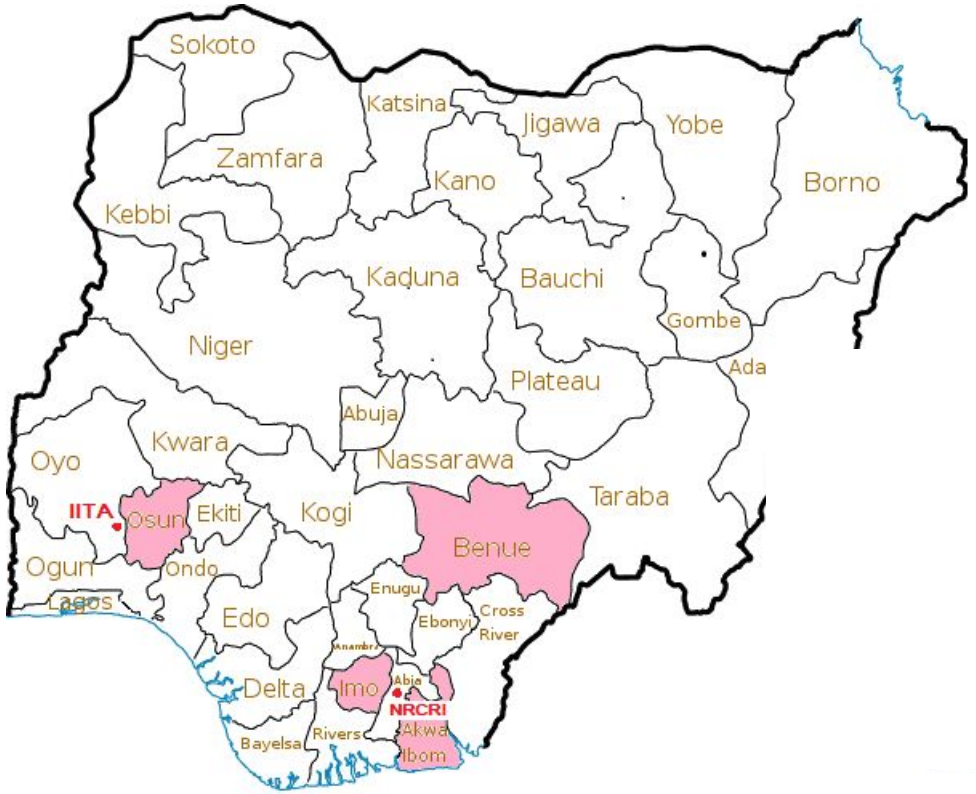
4 states selected

Based on activity in cassava cultivation and processing

2 out of the three senatorial zones per state: ADP consulted to know the 2 most active zones in the state out of three:

4 communities per zone: ADP at zone level consulted to identify 4 communities per zone.

10 farmers per community: purposive sampling on expertise and gender & social groups

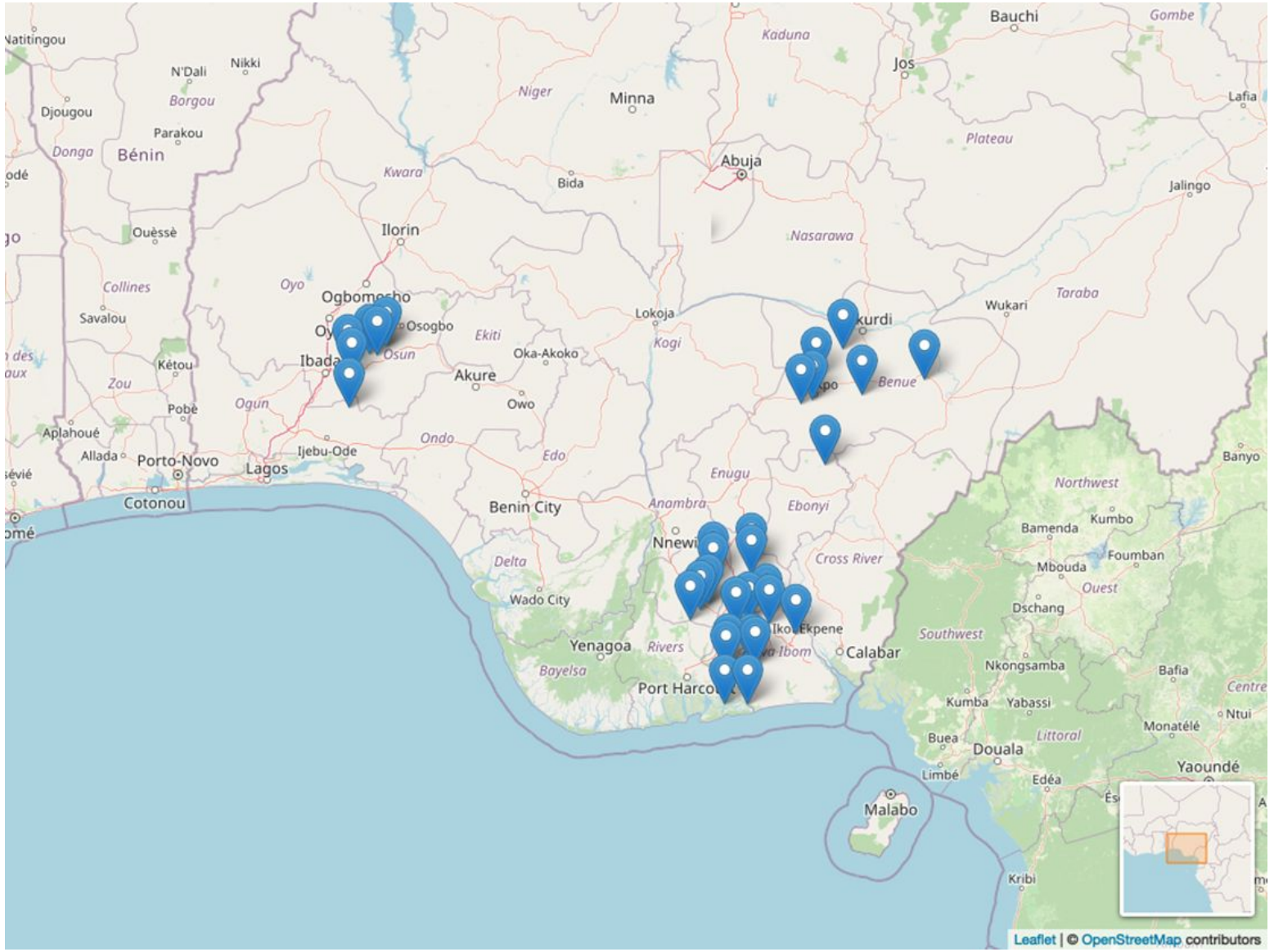


- 80 Tricot trials per state: Total of 320 Trials (5 x 6 plants)
- 30 varieties tested (farmer popular varieties and new ones)
- Each variety features around 32 times

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Tricot citizen science trials in Nigeria

320 participants/trials across 4 States



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Tricot principles

- Balanced Incomplete Block Design (BIBD): Each farmer evaluates **only 3 varieties at different timepoints from planting to processing** while the total nr of varieties can vary (e.g. 8 to 30 or more)
- **Participants are engaged as citizen scientist (co-researchers), results are fed back to them.**
- The number of farmers to include depends on the total nr of varieties that is evaluated
- Tricot evaluation with the farmers **ranks the 3 different varieties** from best to worst:
  - Overall and for specific important traits of the sample
  - No rating involved: concrete comparison between the three varieties (best and worst)

Project design, generating packages of three varieties and data storage and analysis is **centrally managed** on the ClimMob platform
www.ClimMob.net

ClimMob youtube channel:
https://www.youtube.com/channel/UCmqo4KCZwX8R-H4SNkXfuSA

Particular videos to highlight:Design a tricot trial
https://www.youtube.com/watch?v=y3rjXYTrxZI&t=2467s

Registration, Data Collection and Report with ClimMob
https://www.youtube.com/watch?v=y90SjS4hMzs&t=111s

Data collection with ODK Collect
https://www.youtube.com/watch?v=3O3nwH7TqVE

Van Etten et al. (2020) the tricot citizen science approach applied to on-farm variety evaluation: methodological progress and perspectives
https://doi.org/10.4160/23096586RTBWP20212

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TOTAL nr of varieties (farmer popular and newly bred)

VAR 1
VAR 2
VAR 3
VAR 4
VAR 5
VAR 6
VAR 7
VAR 8
VAR 9
VAR 10
……

Different combinations of 3 varieties

Farmer 1
VAR 1
VAR 6
VAR 9

Farmer 2
VAR 4
VAR 2
VAR 3

Farmer 4
VAR 10
VAR 1
VAR 8

Farmer 3
VAR 4
VAR 6
VAR 7

Farmer …
Etc. (*until total nr of farmers*)

(sequentially balanced randomization)

e.g. package 1-5 contain all 10 varieties
Then package 5-11 contain all 10 varieties

Final rank of the varieties for overall + traits

VAR 8
VAR 18
VAR 4
VAR 10
VAR 11
VAR 5
VAR 6
VAR 1
VAR 21
VAR 7
VAR 2
VAR 9
VAR 3
…..

(Plackett-Luce tree model)

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Setup

320 farmers across 4 states, **30** varieties

- Rank overall, and for important traits, best and worst plants and food product made from them, at different stages
- Processing is done and evaluated by the processor of the Tricot farmer's household
- Final choice of the best overall variety after the whole experience is done with all the household members involved
- Compare trends for gender and region and e.g. farm size and other social identities and intersections between these
- Use of RHoMIS social questionnaire with each Tricot participant
- Tricot participants determine metric yield using local measures (to live up to variety release demands)

trail



roots





gari



eba

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Agronomic and post harvest evaluation moments for ranking of the varieties: best and worst of the three in each farmer's field

Before harvest

TIME AFTER PLANTING	Trait/characteristic
1 month	• Germination/Sprouting • Plant growth (vigour, strength) • Suitability to soil and environment • ***Overall impression***
3 months	• Survival/sprouting • Plant growth (vigour, strength) • Weed competitiveness/canopy formation • Branching habit/architecture (shape, height, forms of the plant) • Diseases/Pest resistance (deformation of leaves etc.) • Early bulking • ***Overall impression***
6 months	• Survival/sprouting/number of plants left • Plant growth (vigour, strength) • Weed competitiveness/ canopy formation • Branching habit/architecture (shape, height, forms of the plant) • Disease resistance (deformation of leaves etc.) • Early bulking • ***Overall impression***
9 months	• Plant growth (vigour, strength) • Weed competitiveness/ stay green/leaf retention • Branching habit/architecture of the plant • Suitability to soil and environment • ***Overall impression***

After harvest

STAGE	Trait/characteristic
Harvesting: fresh roots	Yield Density/heaviness/water content (root should be dense/heavy, not watery) Colour of the root flesh (bright white) Root size (Big roots) Root shape (No constrictions) Root rot (Not rotten) Woody filament/crude fiber content (No woody filaments/crude-fiber) ***Overall impression***
Gari processing	Cake/mash/pulp discoloration (should maintain bright colour) Water release during pressing (Fast release of water during pressing) Swelling of pulp while toasting (pulp should swell in frying board/pan) *Gari* yield (large amount of *gari* per unit of fresh roots) Ease of peeling (Easy to peel) ***Overall impression***
Gari quality	Dryness (Dry/crispy *gari*) Colour (brightness: butter colour/white/yellow) Heaviness (dense *gari*) Smoothness (homogenous in particle size: fine but with pronounced 'grains', not powdery) Taste (good taste, like full not bland. Less considering sourness) Woody filament/crude fibre content (Not many woody filaments -"small sticks"- in the *gari*) ***Overall impression***
Eba quality (AFTER PREPARATION)	Swelling of *gari* when prepared into *eba* (Good swelling of *gari* when making *eba*) Mouldability (Should not sack out, sweat water or stick to the hand) Softness Drawability ('Elasticity') Smoothness/ Easiness to swallow (particles of the *gari* nicely merged) Not stickiness to hand and mouth Taste Colour ***Overall impression***
Eba quality (AFTER STORAGE)	Mouldability (Should not sack out, sweat water or stick to the hand) Softness Drawability ('Elasticity') Smoothness/ Easiness to swallow (particles of the *gari* nicely merged) Not stickiness to hand and mouth Taste Colour ***Overall impression***
Final evaluation	***Overall impression*** *Indicate which of the three varieties the farmer and processor want to replant or not to replant (provide reason for each variety)*

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(Source: Teeken et al. 2021 https://doi.org/10.1111/ijfs.14862)

Day 2

Abolore Bello: Varietal effect on product women's processing drudgery**. Session 2.2E 1:30h to 3:00h** (POSTER)

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Selection of Tricot participants

- In each community a focus group discussion is held at community head level to identify the different social groups and the cassava related activities within these groups.
  - Involvement of various ethnic/ migrant groups or other locally relevant groups (Non indigene, indigenes)

- Participants are then proportionally selected to reflect the different social groups in the village considering sex and age.
  - Equal chances of participation of men and women (based on position and scale of involvement in cassava value chains)

- Participants are selected **based on expertise** with regards to cassava farming/processing and motivation, not based on dialectic skills.









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Defining local social groups

Locally relevant social group (farmer group, migrant workers, landless, **ethnic** group, **religious** group)*	Share of local popula-tion (oral or record share) [B1.4a]	Language (record language) [B1.4b]	Is this group associated with a particular livelihood(s) or crop? (if so, record livelihood(s) and crops) [B1.4c]	Which tasks related to [the RTB crop, here cassava] are generally done by men and women		Is this group among the better off group(s)? Yes……1 No…… 2 [B1.4d]	Is this group among the most politically active & influential group(s)? Yes……1 No…… 2 [B1.4e]
				women	man		
i.Ilaje (originally from Ondo state)	1000	Yoruba dialect	Fishing, selling of fish			2	2
ii.Agatu (Immigrants from Benue state)	500	Agatu	Farming (subsitance and and cash crops), farm labourer, Inlcudes cassava	Farming (weeding, harvesting), processing, marketing	Farming (weeding, planting, harvesting), marketing of fresh roots	2	2
iii. Markurdi (Immigrants from Benue state)	100	Tiv/Igede	Farming (subsistence and cash crops0, farm labourer, Includes cassava	Farming (weeding) processing, marketing	Farming. Marketing of fresh roots	2	2
iv.hausa	10	Hausa	Trading, fishing			2	2
v.Cotonou (Immigrants from Benin republic)	50	Fon/Ewe	Farming (subsistence and cash crop) and particularly cultivate vegetable, tomato and pepper.	Farming (planting, weeding), processing, marketing	Farming and occasional processing	2	2

Standard household survey comprised of different key indicators

RHoMIS
Rural Household Multi-Indicator Survey

www.rhomis.org

User instructions

RHoMIS is a questionnaire survey for interviewing smallholder farm households. It was developed for use on digital screens (phone/tablet). The digital package also includes infrastructure for data upload and storage, real time visualisation, rapid indicator calculation, and analysis functionality. The digital version is also available in multiple languages and with additional modules. The questionnaire in this document contains the core questions of RHoMIS, and is intended to be a guide to the digital version.

The questions are designed to be user-friendly and as quick as possible, while still gathering enough information to understand the farm system, livelihoods and food security. The interview should run more like a conversation, and take about 45 minutes. An excessive level of detail is not required. The meaning of the questions and answers should be clear and self-evident, although the language may need to be adapted for the local context.

RHoMIS
Rural Household Multi-Indicator Survey

Key indicators generated



Selection of the lead farmer and training

Lead farmer chosen by 10 farmers in a community that have a tricot trial and together have all the varieties in tricot's incomplete block

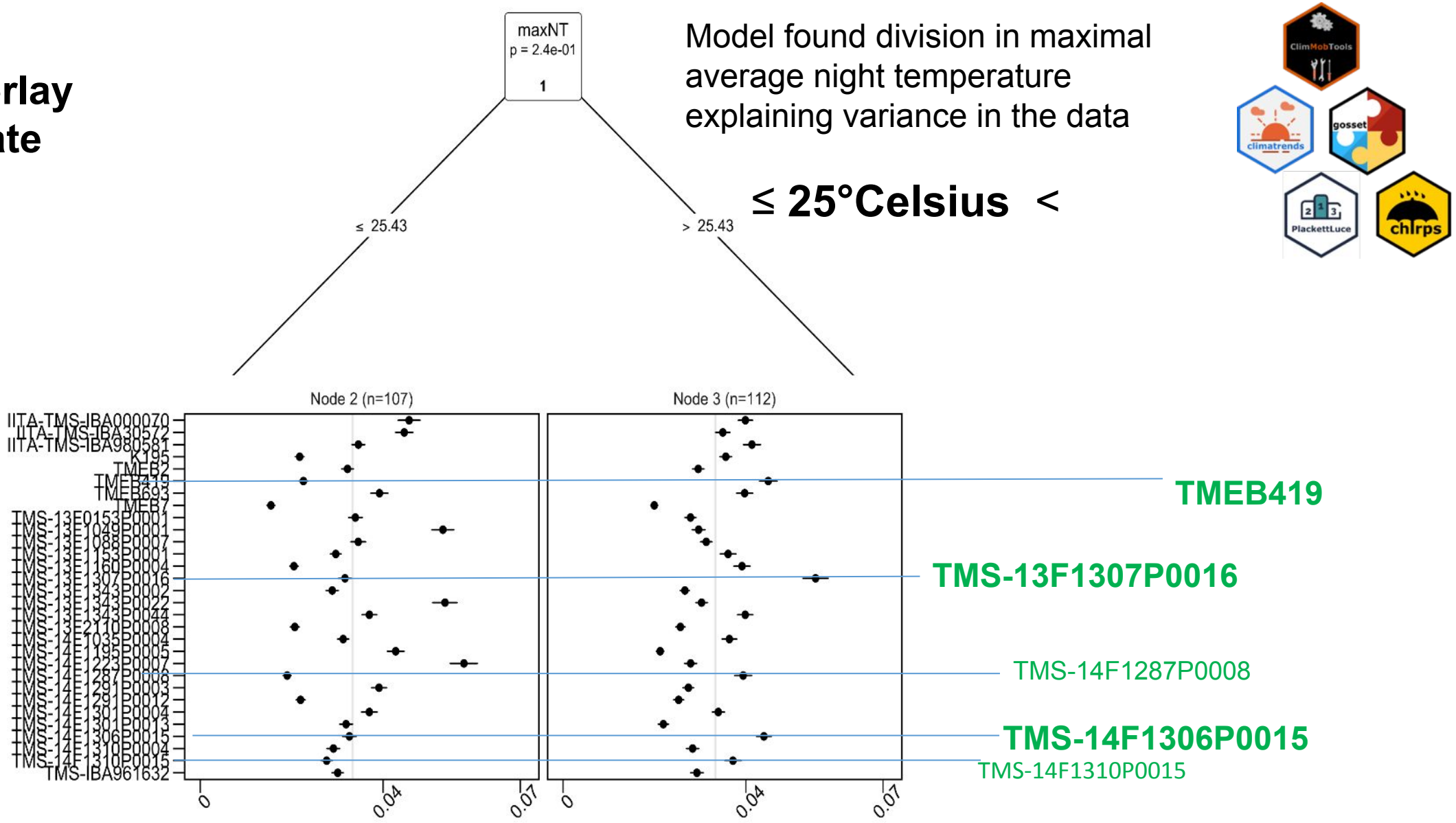
Lead farmer is a respected person in the community and is not made respected because of our project (*Trust, activeness in the community, **Leadership acumen***)



Lead farmer (right) with Tricot farmer in Benue state

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Tricot overlay with climate data



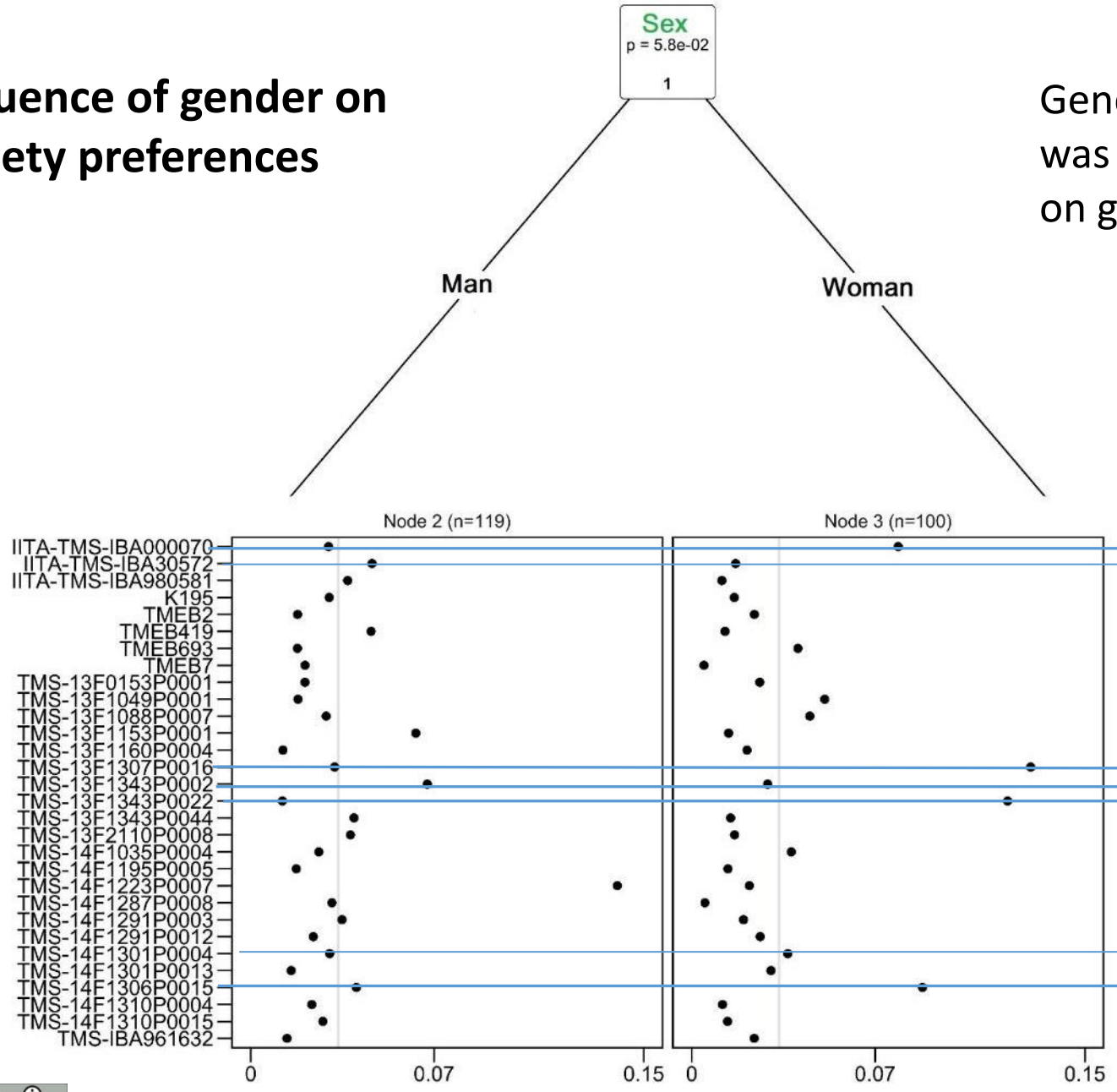
Model found division in maximal average night temperature explaining variance in the data

≤ 25°Celsius <

With the forecast of rising temperatures / climate change this is relevant. **Note good performance of TMEB419 and advanced clones TMS-13F1307P0016, TMS-14F1306P0015 at high night temperatures**

Influence of gender on variety preferences

Gender of the initial listed Tricot participant was most important socio-economic driver on genotype selection



Liked by men

Liked by women

Baba 70 (just released)

TMSIBA30572

TMS13F1307P0016

TMS13F1343P0002

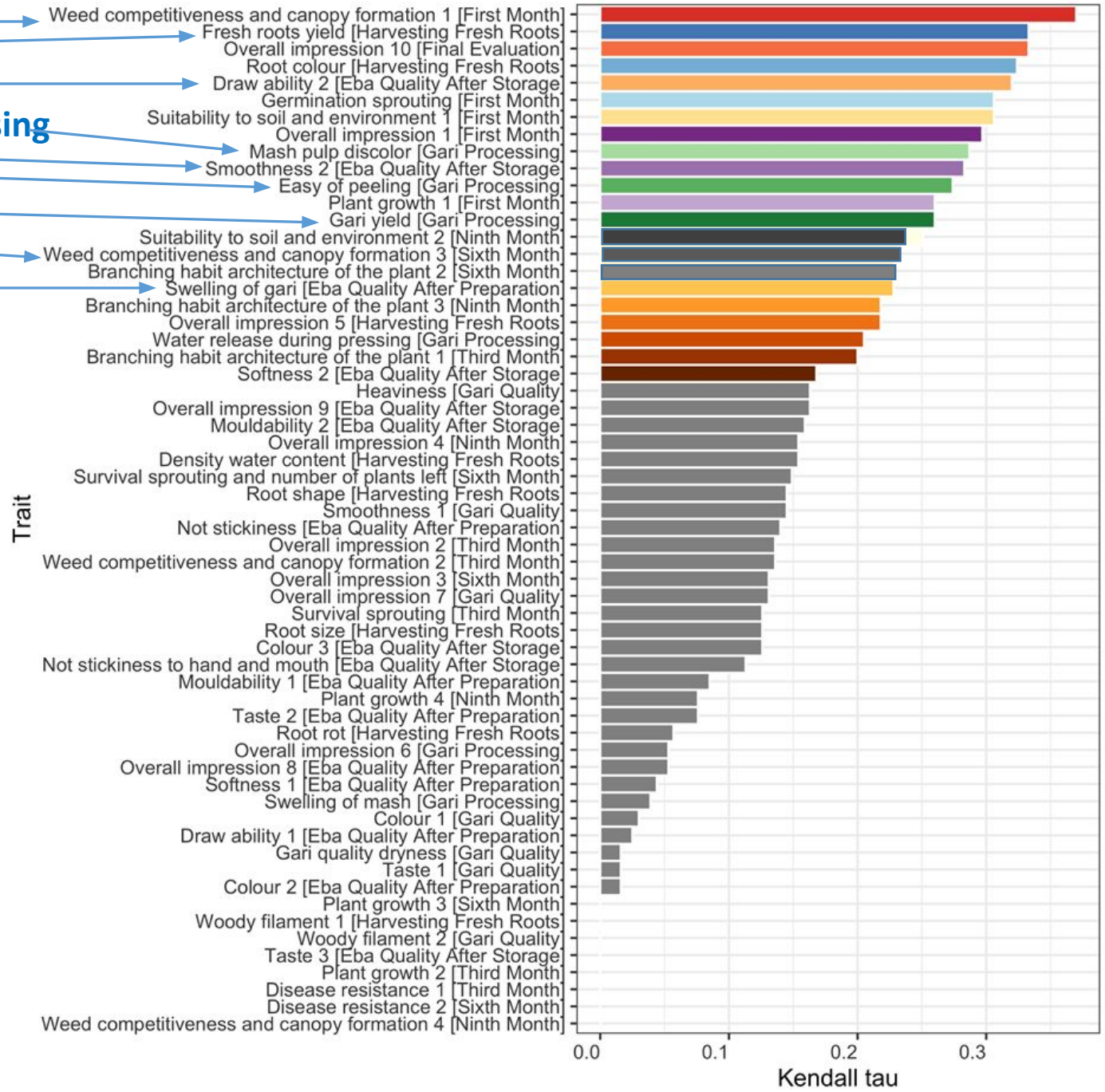
Obasanjo 2 (recently released)

TMS-14F1301P0004

TMS14F1306P0015

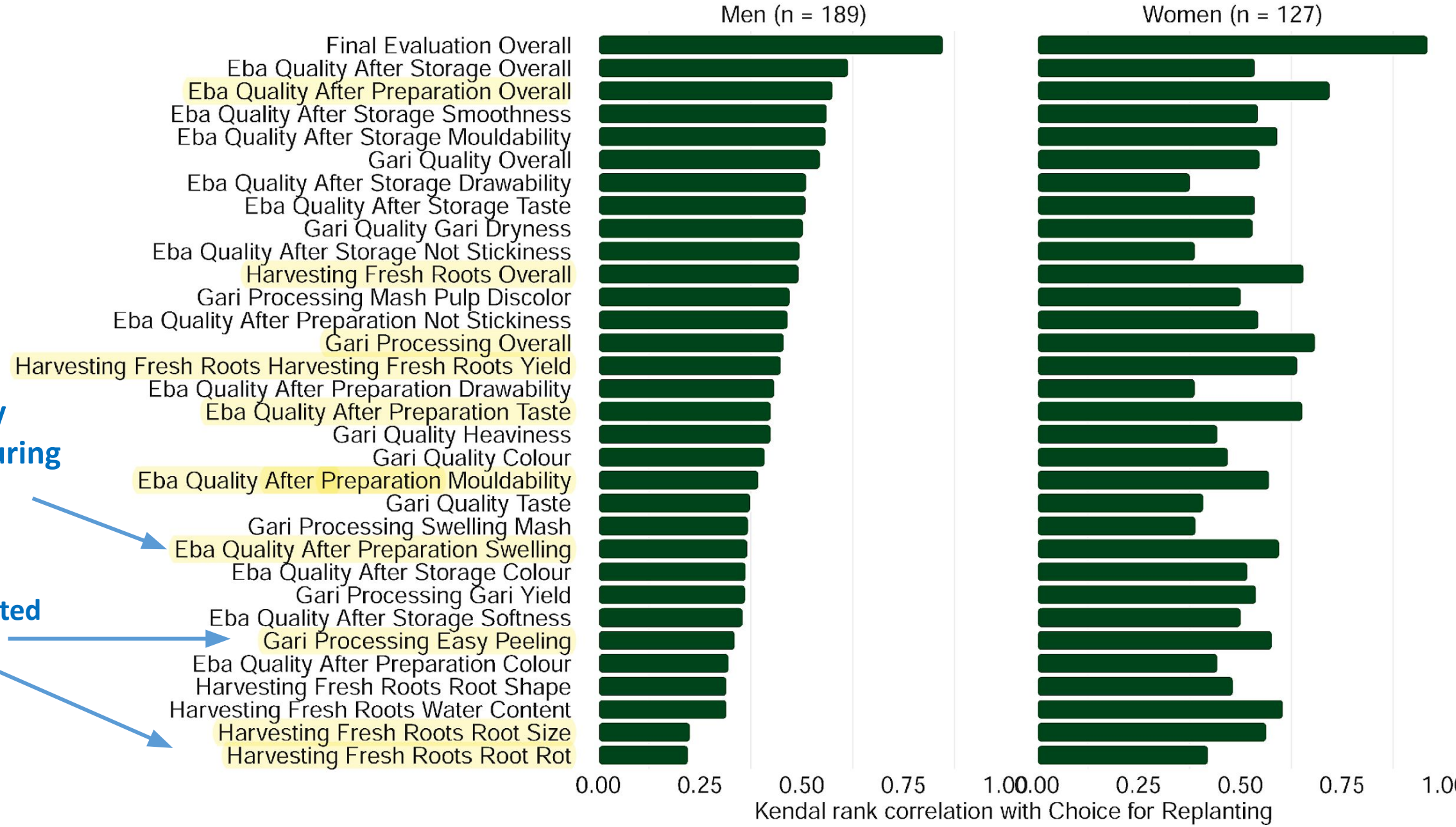
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- Weed competitiveness
- Fresh root yield
- Drawability of eba
- Smoothness eba
- Discoloration during processing
- Ease of peeling
- Gari yield
- Weed competitiveness
- Swelling of gari to eba



Final choice of the best variety after monitoring plants from sowing to food processing:

Highly determined by post harvest traits and **especially by overall food product quality**



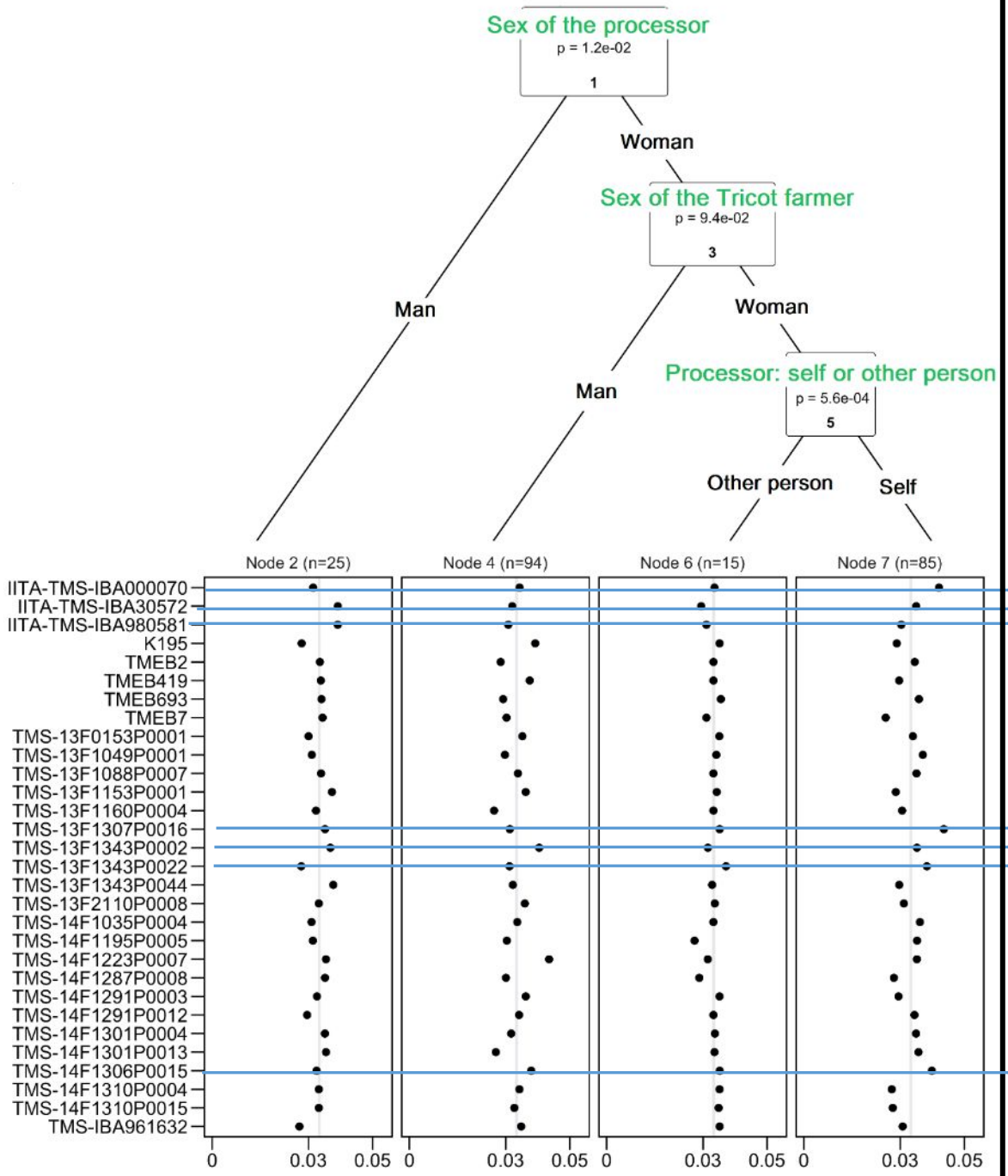
Specifically observe during processing

Peeling related traits

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Looking at who did the processing in the Tricot participant's household.

How does it relate to variety choice and preferences?



Example

Varieties liked by **processors** that are :

Male processor under male Tricot farmer

Female processor under female Tricot farmer

Baba 70 (just released)

TMSIBA30572

TMSIBA980581

TMS13F1307P0016
(Advanced Nextgen clone)

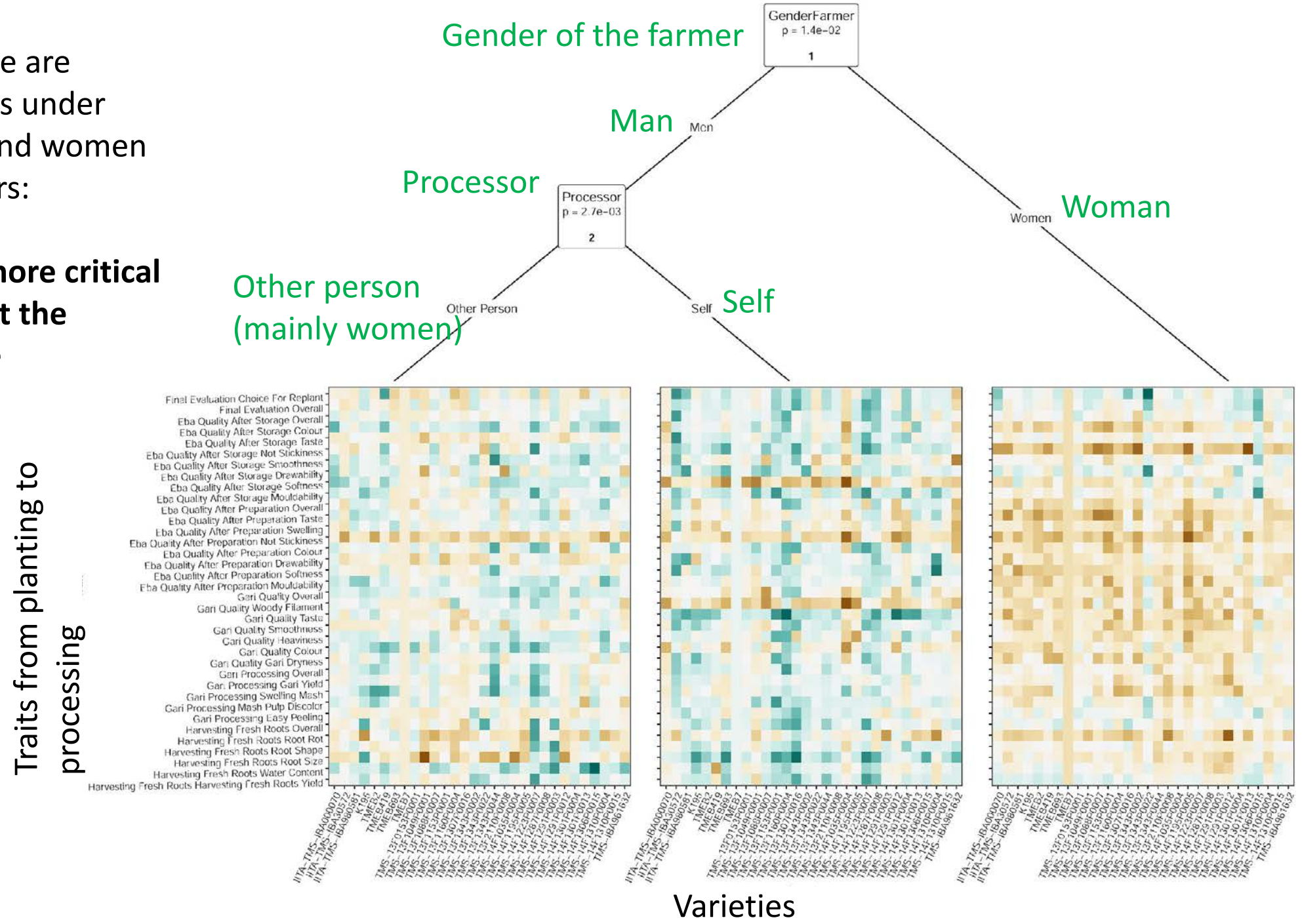
TMS13F1343P0002

Obasanjo 2
(recently released Nextgen clone)

TMS14F1306P0015

There were the are are women processors under women farmers and women under men farmers:

Households are more critical and specific about the varieties they like

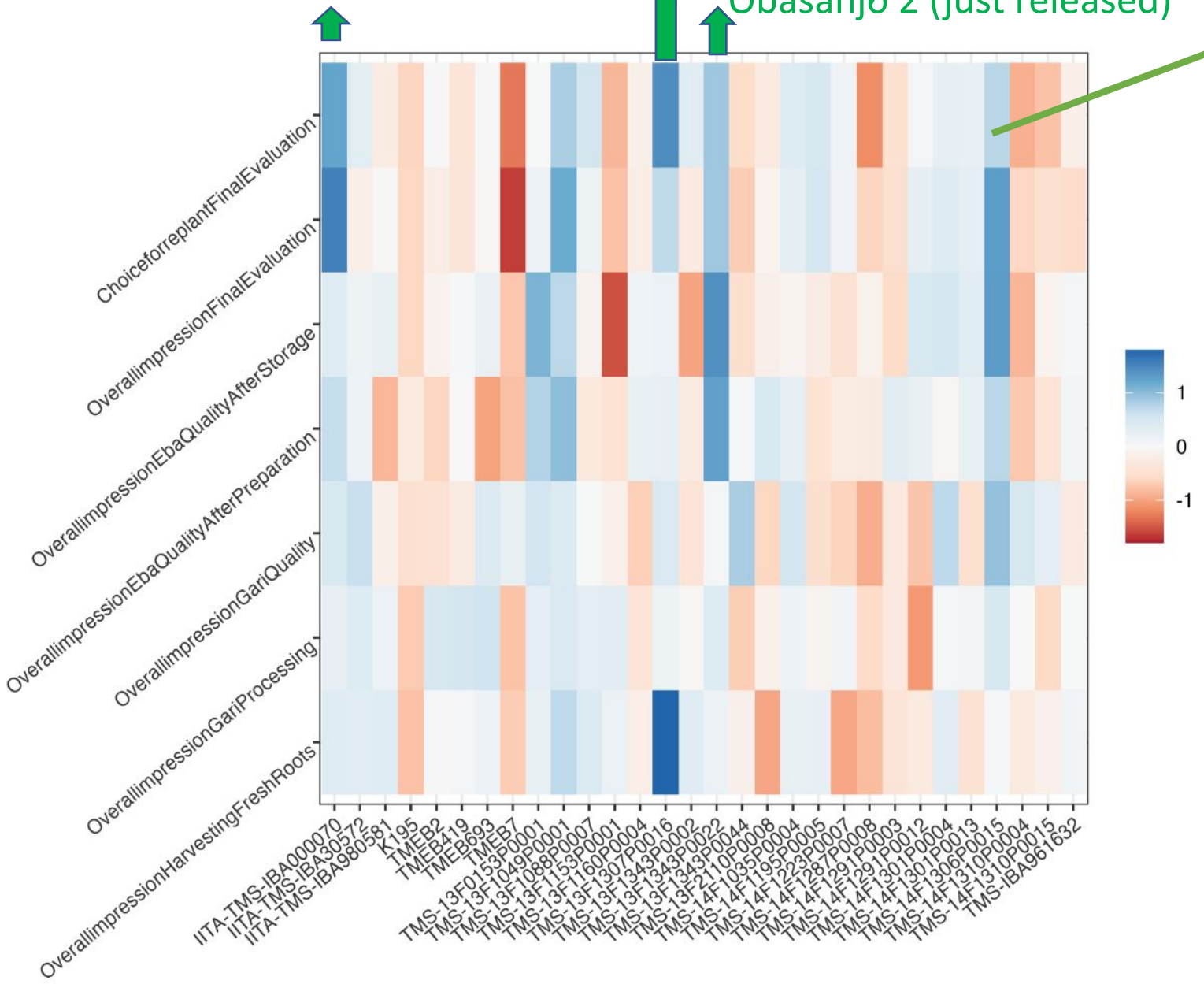


Baba 70 (just released)

TMS13F1307P0016 (advanced)

Obasanjo 2 (just released)

TMS-14F1306P0015 (advanced)



NODE

GenderProcessor = Woman

GenderFarmer = Woman

Processor = Self

Top 5

TMS-13F1307P0016

IITA-TMS-IBA000070

TMS-14F1306P0015

TMS-13F1343P0022

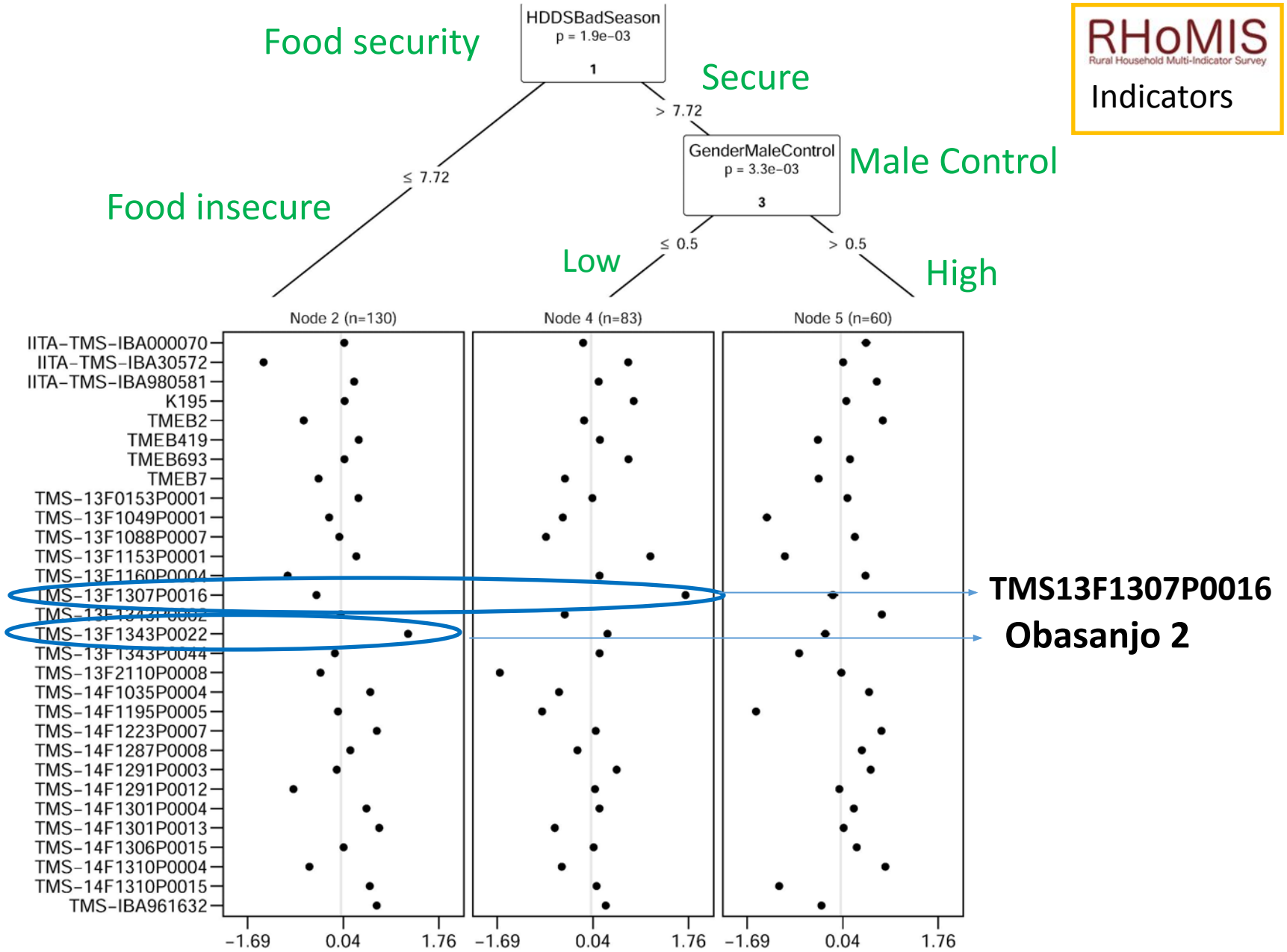
TMS-13F1049P0001

People form food insecure households like particular varieties like the released:

Obasanjo 2

People from food secure household with low male control like the advanced:

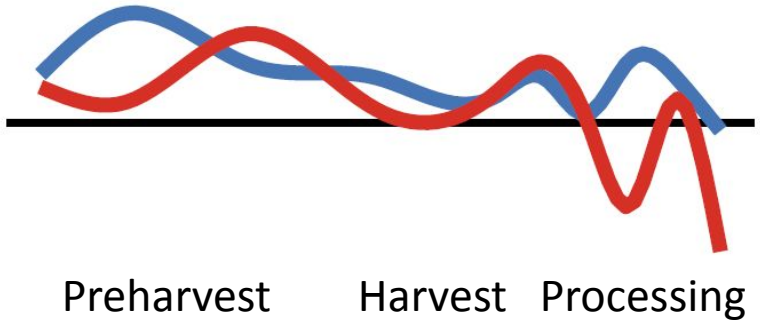
TMS13F1307P0016



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Old released variety liked by men farmers who process themselves (small nr)

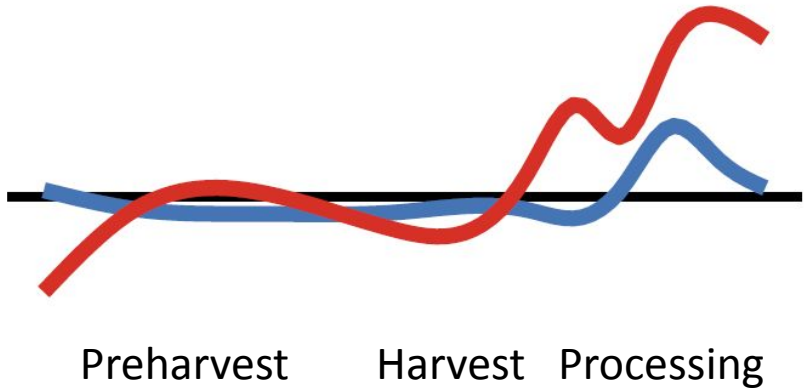
IITA-TMS-IBA980581



Varieties liked by women farmers who process themselves

'Food secure low male control'

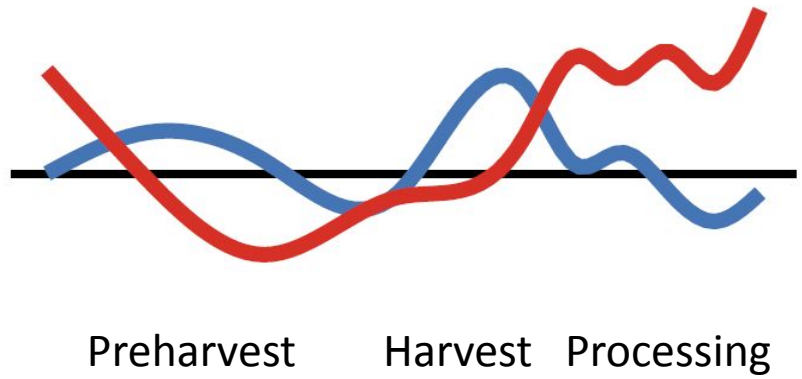
TMS-14F1306P0015



'Food insecure'

TMS-13F1343P0022

(Released as: Obasanjo 2)



! Improvement envisaged:
Evaluate all moments with 2 people (dyads) in the Tricot farmers households

Men Women

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Cooperation with Nigerian variety release Committee on Tricot

- Our Tricot work was noticed by the CSP program https://csp-nigeria.org/ that works in innovating Variety Release

- Tricot was accepted by the Nigerian Variety release Committee to try out as a new way to do on farm testing

- CSP, Cassava breeding unit at IITA see Tricot as more than that: breeding trials that provide earlier feedback from users (before on-farm testing)

- An extensive Tricot setup can easily nest the requirements for variety release

- **Current and next year's Tricot are owned by variety release committee and CSP**





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IITA
Transforming African Agriculture
CGIAR
SAHEL Consulting
COLLABORATIVE SEEDPROGRAMME
NEXTGEN CASSAVA
Tricot: An alternative method for on farm testing for variety release
What to expect
An innovative participatory and scalable way of doing on farm testing
Focus on improved assessment of external validity of varieties
Experiences of Tricot from Uganda, Ghana and Nigeria
SPEAKERS
Dr Sunday Aladele
NACGRAB
Dr Jacob van Etten
Alliance Bioversity-CIAT
Dr Kaue de Sousa
Alliance Bioversity-CIAT / I.N. University of Applied Sciences
Prof. Chiedozie Egesi
IITA / NRCRI / Cornell University
Dr Paula Iragaba
NaCRRI
Dr Ann Ritah Nanyonjo
NaCRRI
Dr Richard Oteng-Frimpong
SARI
Chinedu Agbara
Sahel Consulting
Dr Béla Teeken
IITA
Prof. Olusoji Olufajo
IAR / ABU
Tuesday,
12 April 2022
10:00 AM (WAT)
Alliance
Bioversity
CIAT
1000FARMS
Trial Network

CC BY

Discussing redesign with variety release committee to improve **on farm testing** within the variety release procedures

(NACGRAB Ibadan, Nigeria)

GOAL

After on farm test:

Automated reporting including social segments and intersection with gender

Integration

ClimMob

BREEDBASE



Source: Twitter X https://x.com/BelaTeeken/status/1671990768322453504?s=20

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Tricot in sum

- Systematic storage of on farm data and integration with breeding trial data
- Standard incorporation of social factors and their intersection with gender (target social impact, breeding investment cases)
- Towards measuring external validity of breeding trails: **genetic gain in farmers fields** (donor requirements)
- A platform for more detailed qualitative action gender and social research to better interpret Tricot results
  - Plans to improve tricot on gender: introducing evaluation with dyads for all evaluation moments
- A network that can closes the gap between public breeding and variety users: **users as partners and citizen scientists in the breeding program. I human centered problem focused design approach**
- A network that can link NARES, NGOs, private sector which will make it even more cost effective
- A network open to everyone with expertise in crop work, not demanding deliberation skills: A concrete way to **performatively** rather than deliberatively engage users.

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Béla Teeken, Olamide Olaosebikan, Abolore Bello, Hale Ann Tufan, Martina Occelli, Kaue de Sousa, Tessy Madu, Samuel Onwuka, Jacob van Etten, Gospel Edughaen, Durodola Owoade, Afolabi Agbona, Joost van Heerwaarden, Peter Iluebbey, Elisabeth Parkes, Peter Kulakow



(Source: https://gender-portal.rtb.cgiar.org/farmer-stories/working-together-for-better-gari/)

BILL & MELINDA GATES foundation

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